

MSBA Job Matching

Grounded NLP decision support for career advisors

Real postings -> retrieved evidence -> role-fit triage -> human review

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The advisor must separate fit, uncertainty, and risk

A useful system improves a specific decision: forward, hold, or investigate.

FORWARD

Strong entry-level analytics fit supported by title, skills, and location evidence.

HOLD

Senior, engineering-heavy, or clearly misaligned roles require a conservative screen.

INVESTIGATE

Incomplete fields or missing authorization language require human follow-up.

Role fit and work authorization are related in workflow, but they are not the same prediction task.

Label quality, not data volume, is the bottleneck

The final model uses the full training split and audits all 785,741 source rows.

785,741

original public job
records

100,000

stratified project
sample

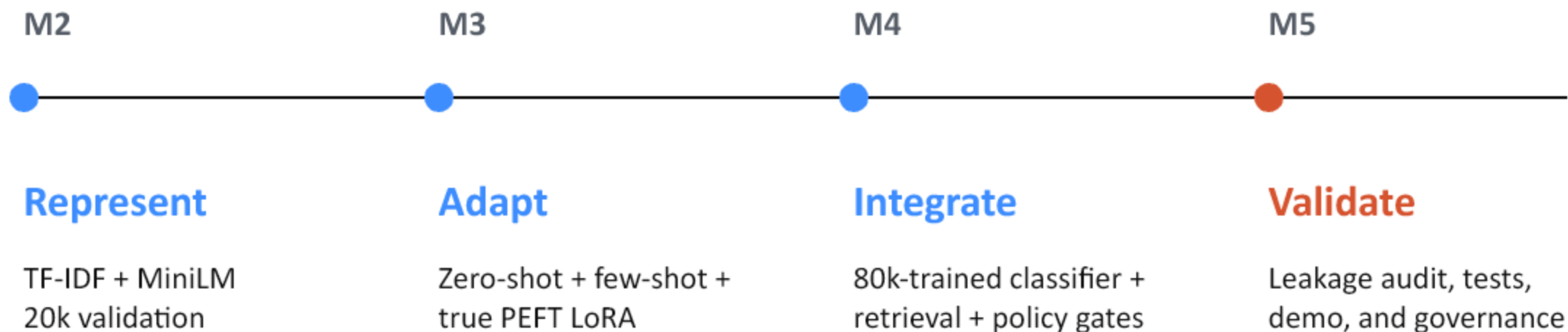
77,135

deduplicated training
rows

230

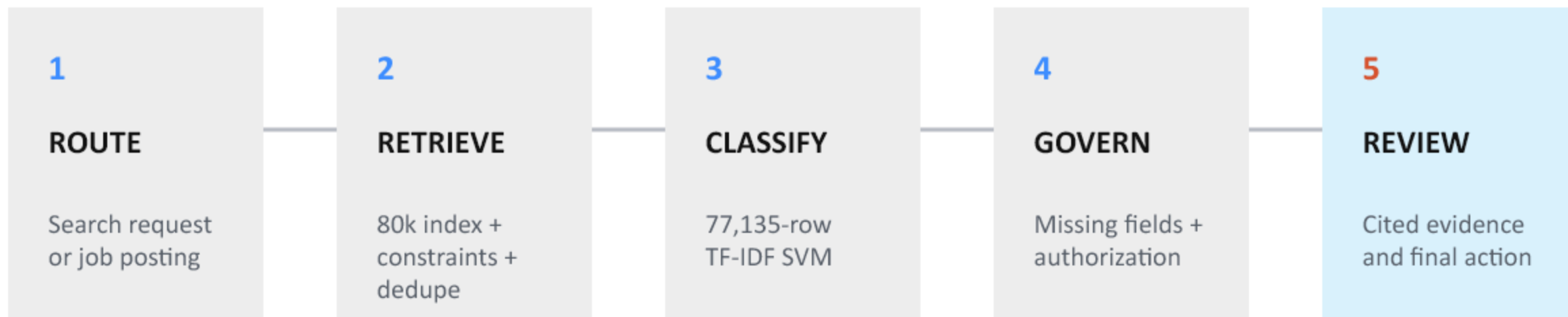
unique authorization
candidates

Each milestone changed the final technical decision



Two input modes prevent a subtle but costly task error

Search requests retrieve candidates; only job postings receive a role-fit label.



A high role-fit score can never override an explicit authorization restriction.

Search mode returns constrained, inspectable candidates

The top five results satisfy the requested US constraint; the search sentence is not misclassified.

INPUT Find entry-level US data analyst, business analyst, or BI roles with SQL, Python, Excel, Tableau, or Power BI. Remote roles are acceptable.

RANK	POSTING	COMPANY / LOCATION	FIT	SCORE
1	Data Analyst	HireMatch Anywhere	high_fit	10.354
2	Data Analyst (US REMOTE)	LeanTaaS Anywhere	high_fit	9.158
3	Data Analyst	HMP Global Anywhere	high_fit	9.055

Review six retrieved postings; the top five satisfy the US constraint.

Authorization: verify separately unless explicit source text is present.

Edge cases change the action, not the safety boundary

ENTRY LEVEL ANALYTICS SEARCH

**search
results**

Find US analytics opportunities for
advisor review

SENIOR ENGINEERING POSTING

low_fit

Hold a role that is too senior and
engineering-heavy

AUTHORIZATION RESTRICTION POSTING

high_fit

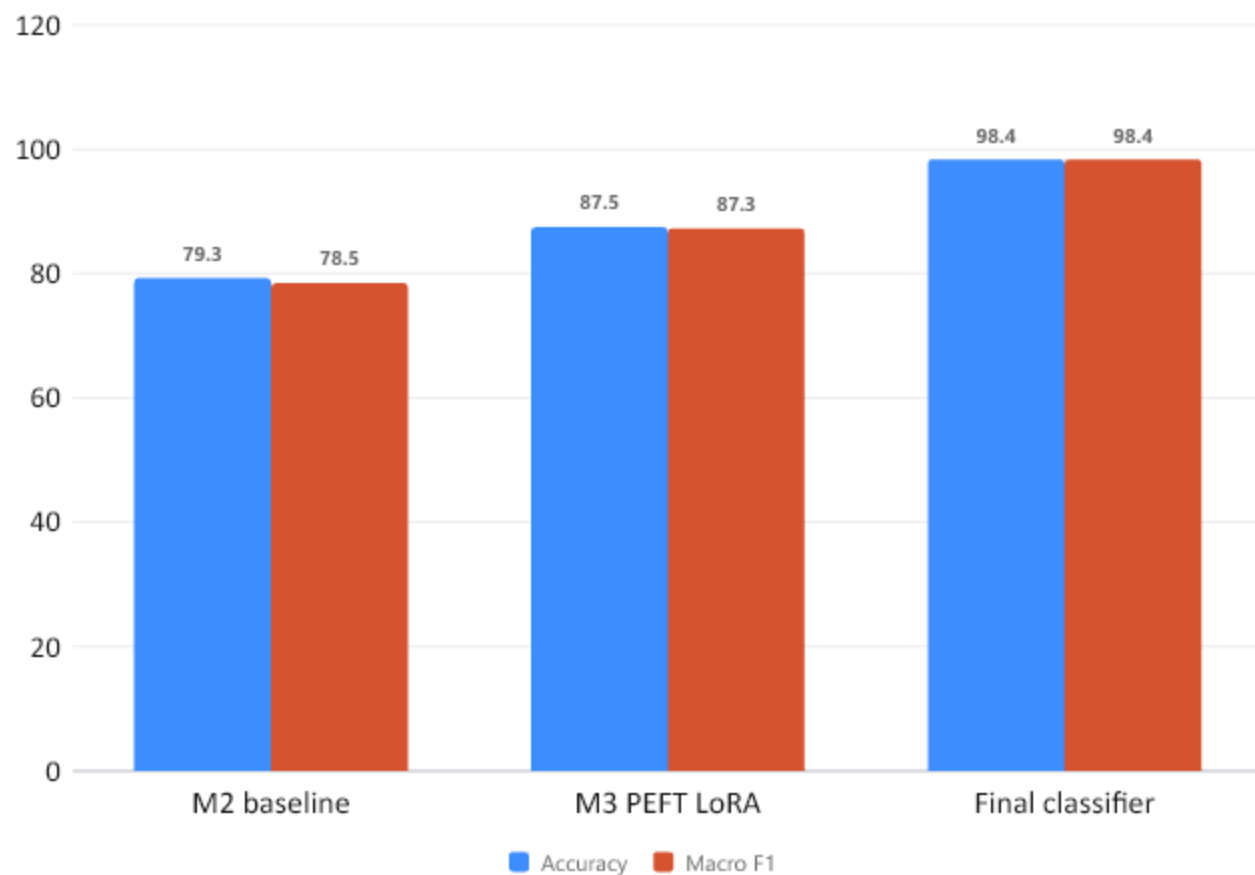
High role fit, but No OPT/CPT forces
Hold

THIN METADATA POSTING

unclear

Missing fields override medium_fit
to unclear

Using all training data beats extra model complexity



98.4% fixed-set macro F1

97.9% novel-text macro F1

0 unsupported
authorization claims

The final 3.8 MB model trains in under one minute.

The remaining risk is label validity, not model capacity

FIXED 4K ERRORS BY TRUE LABEL



DEPLOYMENT RULE

Treat 0.984 as weak-label agreement, not advisor accuracy.

Prepared validation assets:
1,000-row advisor queue
430-row authorization benchmark
Time-split macro F1 0.976

The system and its human-review package are complete

Delivered now

- Trained classifier
- Four-case policy test
- Time-based validation
- Reproducible CLI and tests

PREPARED EVIDENCE

- 1,000-row advisor annotation queue
- 230 unique authorization candidates
- 430-row candidate/control benchmark